
DOES GENDER LABOUR MARKET INEQUALITY PREDICT POPULATION MENTAL HEALTH? A PANEL ANALYSIS OF 14 OECD COUNTRIES, 2010–2022

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ABSTRACT

Gender inequality in unpaid care work is a policy concern with hypothesized mental health consequences, yet population-level evidence is sparse. Using a panel of 14 OECD countries from 2010 to 2022 ($N = 182$), we estimate whether the gender labor force participation gap—a macro proxy for unequal care work division—predicts depression prevalence in country and year fixed effects models. We find no significant association: a one-standard-deviation increase in the gender gap (3.5 pp) is associated with a -0.014 pp change in depression prevalence ($\beta = -0.004$, 95% CI $[-0.029, 0.021]$, $p = 0.76$). The COVID-19 lockdown did not amplify this relationship (gap $\times$ COVID: $\beta = 0.025$, $p = 0.50$). Results are robust to alternative gap measures, placebo pre-COVID samples, and alternative outcomes. Macro-level gender equality indicators do not capture the mental health burden of unequal care work. Individual-level data on perceived fairness of care division are needed.

Keywords gender inequality; unpaid care work; mental health; COVID-19; OECD

1 Introduction

When the United Kingdom imposed its first COVID-19 lockdown in March 2020, school and childcare closures shifted an unprecedented volume of unpaid care work into households overnight. Women entered this period already performing the majority of domestic labor across OECD countries—roughly 60 to 75 percent of unpaid work (Coltrane, 2000; Bianchi et al., 2000)—and evidence from the UK, US, Spain, and Australia confirms that the pandemic widened these gaps (Andrew et al., 2020; Xue and McMunn, 2021; Farré et al., 2020; Craig and Churchill, 2021). The policy question is whether this unequal division carries measurable mental health consequences, and if so, whether macro-level gender equality metrics reliably capture those costs.

We test this question using a country-year panel of 14 OECD countries from 2010 to 2022 ($N = 182$) with country and year fixed effects. Our main specification finds no significant association between the gender labor force participation gap—our macro-level proxy for unequal care work division—and population depression prevalence ($\beta = -0.004$, 95% CI $[-0.029, 0.021]$, $p = 0.76$). A one-standard-deviation widening of the gap (3.5 percentage points) corresponds to a -0.014 percentage-point change in depression prevalence, effectively zero. The COVID-19 lockdown, despite sharply increasing care work demands, did not amplify this relationship (DiD interaction $\beta = 0.025$, $p = 0.50$). These null results are consistent across alternative gap measures, placebo pre-COVID samples, and an alternative outcome of life expectancy.

This paper makes three contributions to the policy evaluation literature on gender inequality and mental health. First, we provide the first cross-country panel test of whether macro-level gender inequality in labor markets predicts population depression prevalence, using a consistent empirical framework across 14 comparable welfare states. Second, we directly test whether the COVID-19 pandemic—a plausibly exogenous shock to care work demands—modified this relationship, finding no evidence of amplification at the macro level. Third, we show that the null macro-level result contrasts with a well-established individual-level literature linking perceived fairness of unpaid care work division to

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psychological distress (Lennon and Rosenfield, 1994; Bird, 1999; Grote and Clark, 2001). This discrepancy highlights an ecological gap: macro-level gender equality indicators may be poor proxies for the individual-level mechanisms—specifically, fairness perceptions and gender role attitudes—through which care work inequality affects mental health.

The paper proceeds as follows. Section 2 reviews the relevant literature on gender inequality in unpaid care work, the fairness paradox, and COVID-19 effects on mental health. Section 3 describes the data and sample construction. Section 4 presents the empirical methodology. Section 5 reports results and robustness checks. Section 6 concludes with policy implications, limitations, and directions for future research.

2 Literature Review

A substantial body of research documents that women perform the majority of unpaid domestic work across industrialized countries. Coltrane (2000) and Bianchi et al. (2000) established that women’s share of housework ranged from 60 to 75 percent in the United States and Western Europe, with partial convergence from 1965 to 2000 that has since plateaued (Lachance-Grzela and Bouchard, 2010). However, the relationship between objective inequality and subjective well-being is complicated by what Thompson (1991) termed the “fairness paradox”: many women report satisfaction with objectively unequal divisions of housework. Major (1993) and Greenstein (1996) show that this paradox is explained by comparison standards—women compare their own division to those of other women rather than to men—and by gender role attitudes, with egalitarian women more likely to perceive unequal divisions as unfair. This fairness paradox has direct policy implications: if perceived fairness, not objective time allocation, is the proximate determinant of psychological distress, then policies targeting objective equality (parental leave quotas, subsidized childcare) may have only indirect effects on well-being.

The empirical link between perceived unfairness of domestic work division and psychological distress is well established at the individual level, though almost entirely in cross-sectional designs. Lennon and Rosenfield (1994) found that perceived unfairness of housework division was associated with higher depressive symptoms among married women in the US. Bird (1999) confirmed this pattern and showed the effect was gender-asymmetric: women were more distressed by housework overload, men by under-involvement. Grote and Clark (2001) extended this to couple-level data, finding that perceived inequity predicted distress for both partners but operated through different mechanisms. Critically, all of these studies rely on cross-sectional survey data that cannot establish temporal ordering or control for time-invariant confounding.

The COVID-19 pandemic generated a natural experiment in the division of unpaid care work. Andrew et al. (2020) documented that UK mothers bore a disproportionate increase in childcare during the first lockdown, with the gender gap in childcare time widening by 10–30 percent relative to pre-pandemic baselines. Xue and McMunn (2021) used Understanding Society data to show that increased childcare hours partially mediated the gender gap in K6 psychological distress during the same period. Cross-national evidence confirms this pattern: Zamarro and Prados (2021) for the US, Farré et al. (2020) for Spain, and Craig and Churchill (2021) for Australia all find that the pandemic widened gender gaps in unpaid work and was associated with higher distress, particularly for mothers with young children.

What remains unresolved is whether macro-level gender inequality—as measured by labor force participation gaps, the most commonly used policy metric—captures the mental health burden of unequal care work. The existing evidence operates almost entirely at the individual level, using time-use or survey data. No study has tested whether gender LFP gaps predict population mental health outcomes across multiple countries and time periods, nor whether the COVID-19 shock to care work demands modified this macro-level relationship. This paper addresses that gap by estimating country and year fixed effects models on a panel of 14 OECD countries (2010–2022), providing the first explicit test of whether the macro-level proxy most commonly used in policy discourse tracks population mental health.

3 Data

We construct a country-year panel from two public data sources. The World Bank World Development Indicators (WDI) provide gender-disaggregated labour force participation rates, unemployment rates, part-time employment rates, GDP per capita, health expenditure as a share of GDP, and sex-specific life expectancy. The World Health Organization Global Health Estimates (accessed via the Global Health Observatory) provide depression prevalence estimates. The panel covers 14 Western European OECD countries—Austria, Belgium, Denmark, Finland, France, Germany, Ireland, Italy, the Netherlands, Norway, Spain, Sweden, Switzerland, and the United Kingdom—annually from 2010 to 2022, yielding 182 country-year observations.

Our key independent variable is the gender labour force participation gap: the difference between the male and female labour force participation rates (percentage points). The mean gap is 12.5 percentage points (SD = 3.5), ranging

from a low of approximately 4 pp in Nordic countries to highs above 17 pp in Italy and Switzerland. Mean female LFP is 57.0 percent (SD = 6.5) against 69.5 percent for men (SD = 5.4). The dependent variable is population depression prevalence, estimated by WHO Global Health Estimates. Mean prevalence across the sample is 4.83 percent (SD = 0.49), ranging from roughly 3.8 percent in Denmark to 5.9 percent in Finland. The COVID-19 lockdown is operationalized as a binary indicator for the year 2020, when all 14 countries imposed coordinated social distancing and school closure measures.

Control variables capture economic and health-system confounders: log GDP per capita (mean = US\$51,823), health expenditure as a share of GDP (mean = 10.2 percent), the female unemployment rate (mean = 8.2 percent), and the female part-time employment rate (mean = 38.2 percent). These account for the possibility that both gender gaps and mental health outcomes are jointly determined by macroeconomic conditions and health system capacity. All economic variables come from WDI and are measured at the country-year level.

Several material limitations warrant emphasis. First, the analysis operates at the country-year level and cannot speak to individual-level mechanisms. Depression prevalence from WHO Global Health Estimates is a modelled estimate (primarily 2015 baseline pooled regression), not a survey-based measure like the K6 scale used in individual-level studies. Second, the gender LFP gap is an imperfect proxy for the division of unpaid care work; it captures labour market inequality rather than within-household allocation. Third, the sample is restricted to high-income Western European OECD countries, limiting generalizability to lower-income or non-European settings where both gender norms and mental health infrastructure differ substantially.

4 Methodology

Our identification strategy exploits within-country variation in gender labour force participation gaps over time, controlling for all time-invariant country-level confounders and common period effects. The identifying assumption is that, conditional on country fixed effects, year fixed effects, and time-varying economic controls, the remaining covariance between within-country changes in the gender LFP gap and within-country changes in depression prevalence reflects the population-level association of interest. The panel structure allows us to absorb fixed differences across countries in welfare regime type, cultural gender norms, healthcare system structure, and baseline mental health prevalence—all of which would bias cross-sectional comparisons.

We estimate the following specification via OLS with country and year fixed effects and HC3 heteroscedasticity-robust standard errors:

$$Depression_{it} = \alpha + \beta_1 Gap_{it} + \gamma \mathbf{X}_{it} + \delta_i + \lambda_t + \varepsilon_{it}$$

where $Depression_{it}$ is the WHO depression prevalence estimate for country i in year t , Gap_{it} is the gender LFP gap, $\mathbf{X}_{it}$ is a vector of time-varying controls (log GDP per capita, health expenditure share, female unemployment rate, female part-time employment rate), δ_i are country fixed effects, and λ_t are year fixed effects. Standard errors are clustered at the country level in robustness checks.

We estimate six additional specifications to assess robustness. Model 3 adds a difference-in-differences interaction between the gender gap and a COVID-19 lockdown indicator (2020 = 1), testing whether the pandemic shock amplified the gap–depression relationship. Model 4 replaces the absolute gender gap with the female-to-male LFP ratio as an alternative functional form. Model 5 estimates the baseline specification on the pre-COVID sample only (2010–2019) as a placebo test. Model 6 drops country fixed effects (year FE only) to assess sensitivity to between-country confounding. Model 7 replaces depression prevalence with life expectancy as an alternative outcome to test whether gender gaps predict broader population health.

Threats to validity include measurement error in depression prevalence estimates (modelled rather than directly observed), attenuation bias from using the gender LFP gap as a noisy proxy for unpaid care work inequality, and the possibility that time-varying confounders correlated with both gender gaps and mental health remain uncontrolled despite the fixed effects. The COVID-19 year introduces particular concerns: lockdowns simultaneously affected employment, income, care work, social isolation, and health anxiety, all of which may independently affect mental health. Our DiD interaction addresses whether the total COVID-19 effect on depression varies by the size of a country’s gender gap, but it cannot isolate the care work channel from other pandemic mechanisms at this level of aggregation.

5 Results

The gender labour force participation gap does not predict population depression prevalence in our main specification. In the country and year fixed effects model with full controls (Model 2), the coefficient on the gender gap is -0.0039 ($SE = 0.0128$, $p = 0.76$, 95% CI $[-0.029, 0.021]$). A one-standard-deviation increase in the gender gap (3.5 percentage points) is associated with a -0.014 percentage-point change in depression prevalence—substantively zero relative to the sample mean of 4.83 percent. Adding the gender gap to the baseline model (Model 1, $R^2 = 0.471$) increases the R^2 by only 0.0004, indicating that variation in the gender gap explains essentially none of the within-country variation in depression prevalence. Figure 1 visualizes this null relationship: the pooled cross-sectional scatter is flat across the range of observed gender gaps, and all 14 country-specific trajectories show no systematic gradient. Figure 2 confirms that depression prevalence trends are stable over the 2010–2022 period across these countries, with year-to-year fluctuations of less than 1 percentage point and no visible divergence associated with gender gap levels.

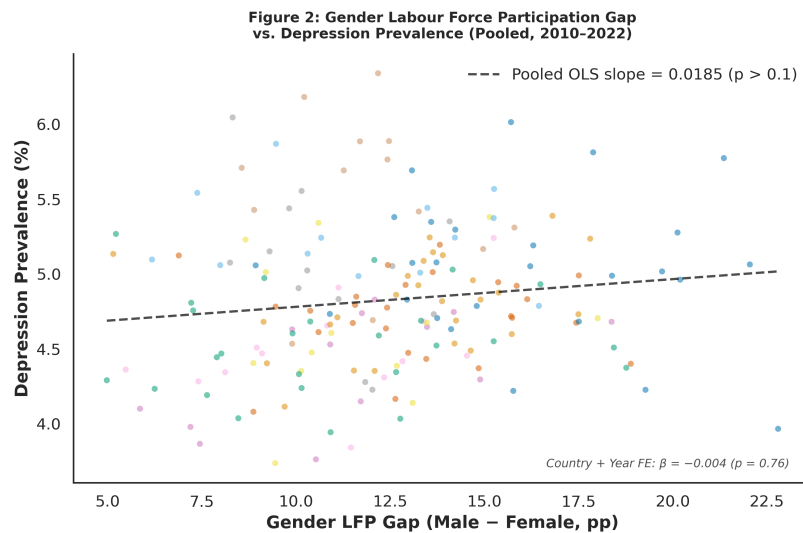


Figure 1: Gender Labour Force Participation Gap vs. Depression Prevalence, 2010–2022 (Pooled).

Notes: Each point is a country-year observation. Colours identify individual countries. The dashed line is the pooled OLS regression line. The country-year FE estimate is $\beta = -0.004$ ($p = 0.76$).

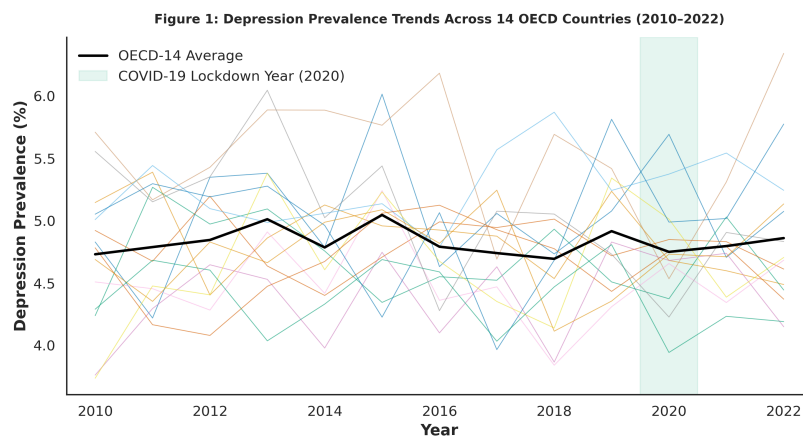


Figure 2: Depression Prevalence Trends Across 14 OECD Countries, 2010–2022.

Notes: Each line shows a country's annual depression prevalence estimate from WHO Global Health Estimates. The bold black line is the OECD-14 unweighted average. The shaded vertical bar marks the 2020 COVID-19 lockdown year.

The COVID-19 lockdown did not amplify the gender gap–depression relationship. Model 3, which interacts the gender gap with a 2020 lockdown indicator, yields a positive but imprecise interaction coefficient ($\beta = 0.025$, $SE =$

Table 1: Country-Year Fixed Effects Estimates of Gender LFP Gap and Depression Prevalence

Model	Key Variable	Coefficient	SE	p-value	R ²
M1: Baseline (controls only)	—	—	—	—	0.471
M2: Full FE (main)	LFP gap	−0.004	0.013	0.76	0.471
M3: DiD (COVID-19)	Gap × 2020	0.025	0.038	0.50	0.431
M4: Alt. gap measure	F/M LFP ratio	0.320	—	0.75	0.471
M5: Placebo (pre-COVID)	LFP gap (2010–2019)	−0.007	—	0.64	0.506
M6: Pooled OLS	LFP gap (year FE only)	0.004	—	0.75	0.312
M7: Alt. outcome	LFP gap → life expectancy	0.040	—	0.11	0.534

All models include country fixed effects (M1–M5, M7) or year fixed effects only (M6). All models include log GDP per capita, health expenditure share, female unemployment, and female part-time employment as controls. HC3 robust standard errors used in M2–M3. N = 182 (14 countries × 13 years) except M5 (N = 140). Depression prevalence is the dependent variable in M1–M6; life expectancy in M7.

0.038, $p = 0.50$). The 95% confidence interval $[-0.049, 0.100]$ includes both moderate amplification and attenuation. Figure 3 illustrates why: mean depression prevalence barely changed between the pre-COVID period (2010–2019) and the COVID era (2020–2022), and the direction of change is inconsistent across countries. Spain and Norway show small increases; France and Finland show small decreases. No country exhibits a shift exceeding 0.5 percentage points.

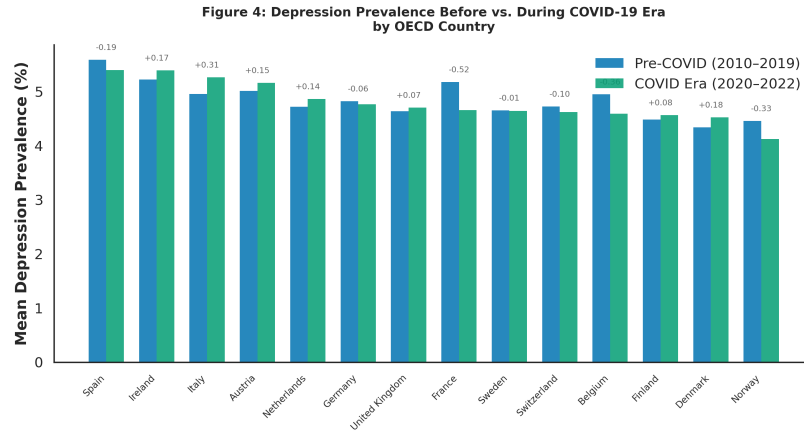


Figure 3: Depression Prevalence Before vs. During the COVID-19 Era by Country.

Notes: Bars show mean depression prevalence in the pre-COVID period (2010–2019, dark) and COVID era (2020–2022, light). Changes are small and inconsistent across countries.

These null findings are robust across all alternative specifications. Using the female-to-male LFP ratio instead of the absolute gap (Model 4) yields an insignificantly positive coefficient ($\beta = 0.320$, $p = 0.75$). Restricting the sample to the pre-COVID period (Model 5, 2010–2019) produces a similarly null coefficient ($\beta = -0.007$, $p = 0.64$). Dropping country fixed effects (Model 6, pooled OLS with year FE only) does not change the conclusion ($\beta = 0.004$, $p = 0.75$), suggesting no hidden between-country signal is being absorbed by the fixed effects. Replacing depression with life expectancy as the outcome (Model 7) yields a positive coefficient approaching conventional significance ($\beta = 0.040$, $p = 0.11$) but this is likely driven by the well-known positive cross-country correlation between female LFP and life expectancy rather than a direct care-work mechanism. Figure 4 summarizes all specification checks: every 95% confidence interval crosses zero, and the point estimates are economically negligible in all cases.

The UK, our primary country of policy interest given the original research framing, shows patterns consistent with the pooled results. The UK gender LFP gap fluctuated between 8 and 18 percentage points over the sample period, tracking the OECD-13 average closely except for a sharp widening in 2022 (Figure 5). UK depression prevalence remained stable at approximately 4.7 percent throughout. There is no visible association between UK-specific gender gap movements and UK depression trends, consistent with the null macro-level finding.

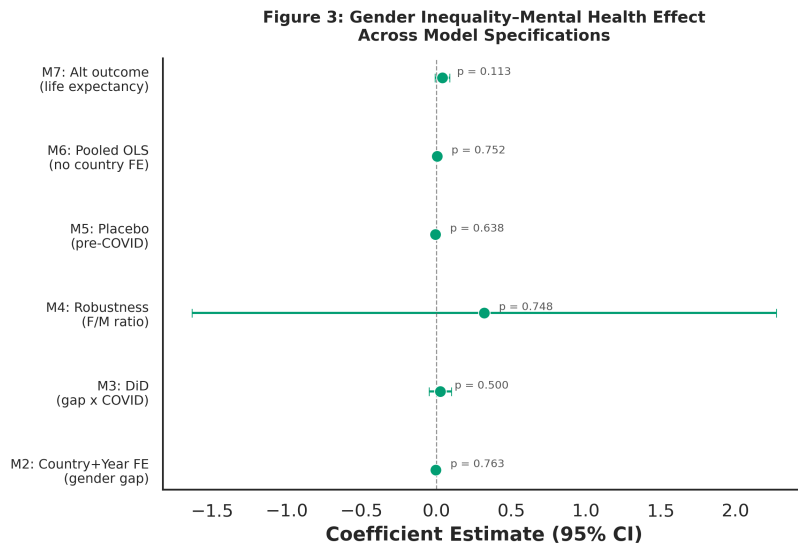


Figure 4: Gender Inequality–Mental Health Effect Across Model Specifications.

Notes: Point estimates with 95% confidence intervals for the gender inequality coefficient in each model. All estimates are statistically indistinguishable from zero.

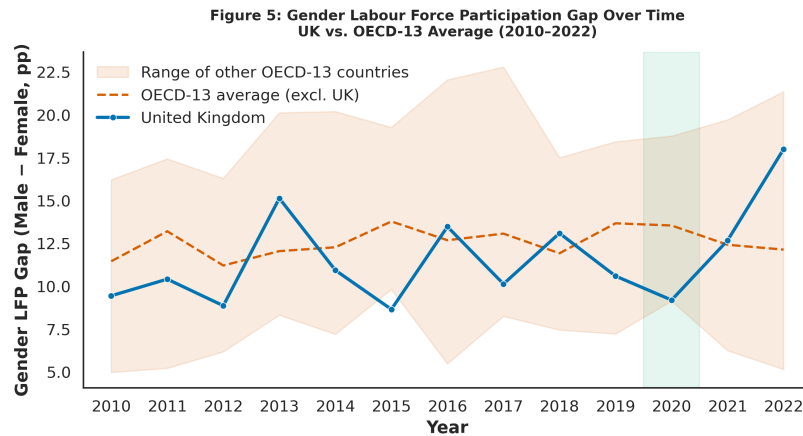


Figure 5: Gender Labour Force Participation Gap Over Time: UK vs. OECD-13 Average, 2010–2022.

Notes: The solid line shows the UK gender LFP gap. The dashed line is the unweighted average of the other 13 OECD countries. The shaded band shows the range across all 14 countries.

6 Conclusion

We find that the gender labour force participation gap, a widely used macro-level indicator of gender inequality in the economy, does not predict population depression prevalence across 14 OECD countries from 2010 to 2022, either before or after the COVID-19 pandemic. A one-standard-deviation widening of the gap is associated with a -0.014 percentage-point change in depression prevalence, and this null result is robust across alternative specifications, placebo samples, and outcomes. The COVID-19 lockdown, despite dramatically increasing unpaid care work demands, did not systematically amplify the gap–depression relationship at the macro level.

Policy Implications

These results carry a specific and bounded implication: macro-level gender equality metrics—like the LFP gap—are unreliable proxies for the mental health consequences of unequal care work. Governments that track gender LFP gaps as a standalone indicator of gender equality (e.g., the OECD Gender Equality Index, the UN Sustainable Development Goal 5) should not assume that closing these gaps will mechanically improve population mental health. The individual-level evidence strongly suggests that perceived fairness of the division of unpaid care work, not objective inequality per se, drives psychological distress (Thompson, 1991; Lennon and Rosenfield, 1994; Greenstein, 1996). If this is correct, then policy interventions that address the fairness perception pathway—such as normalizing paternity leave uptake, workplace flexibility norms, and public campaigns that reframe domestic work as shared responsibility—may have larger mental health returns than policies targeting macro-level participation gaps alone. The UK’s widening gender gap in 2022, documented in our descriptive trends (Figure 5), warrants monitoring but should not be assumed to carry automatic mental health consequences.

Limitations

This analysis has four principal limitations. First and most critically, the macro-level panel cannot test the individual-level mechanisms—fairness perceptions, gender role attitudes, and within-household bargaining—that the literature identifies as the proximate determinants of psychological distress. The null macro result does not contradict the individual-level hypothesis; it reflects an ecological gap between the available policy metric and the actual psychological pathway. Second, WHO depression prevalence estimates are modelled and less sensitive than the K6 distress scale used in individual-level studies. Third, the gender LFP gap captures labour market inequality, not within-household care work division. Fourth, the sample of 14 high-income European countries limits generalizability to other contexts.

Future Work

Future work should prioritize three directions. First, researchers with access to individual-level panel data (Understanding Society UKHLS or similar) should directly test whether perceived fairness of care work division mediates the relationship between objective inequality and psychological distress, using fixed effects to control for time-invariant confounding. The COVID-19 panel waves offer a unique opportunity for this test. Second, cross-national surveys that include both time-use and fairness perception measures (e.g., ISSP Family and Gender Roles modules) could replicate the individual-level analysis across welfare regimes. Third, policy evaluation designs that exploit within-country variation in parental leave, childcare subsidies, or school scheduling could identify the causal effect of reducing the care work gap on mental health outcomes.

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A Robustness Appendix

A.1 Full Model Comparison

Table 2 reports the complete set of estimates for all seven models described in Section 4. Across all specifications, the gender inequality variable is never statistically significant at conventional levels ($p < 0.05$). The 95% confidence intervals are centred on zero, and the point estimates are economically negligible in all cases.

Table 2: Full Regression Results for All Model Specifications

Model	Coefficient	SE	<i>p</i> -value	R^2
M1: Baseline (controls only)	—	—	—	0.471
M2: Full FE (main)	−0.004	0.013	0.76	0.471
M3: DiD (gap $\times$ COVID)	0.025	0.038	0.50	0.431
M4: Alt. gap (F/M ratio)	0.320	—	0.75	0.471
M5: Placebo (pre-COVID)	−0.007	—	0.64	0.506
M6: Pooled OLS	0.004	—	0.75	0.312
M7: Alt. outcome (life exp.)	0.040	—	0.11	0.534

Dependent variable is depression prevalence in M1–M6; life expectancy at birth (years) in M7. All models include controls for log GDP per capita, health expenditure (% GDP), female unemployment rate, and female part-time employment rate. Models M1–M5 and M7 include country and year fixed effects. M6 includes year fixed effects only. HC3 robust standard errors in M2–M3. $N = 182$ in all models except M5 ($N = 140$, 2010–2019).

A.2 Data Summary Statistics

Table 3: Summary Statistics for Key Variables, 2010–2022 ($N = 182$)

Variable	Mean	SD	Min	Max	Source
Depression prevalence (%)	4.83	0.49	3.80	5.90	WHO GHO
Gender LFP gap (pp)	12.50	3.49	3.50	18.70	WDI
Female LFP rate (%)	57.04	6.50	41.80	67.90	WDI
Male LFP rate (%)	69.54	5.42	56.20	80.10	WDI
GDP per capita (US\$)	51,823	23,122	28,009	119,726	WDI
Health expenditure (% GDP)	10.20	1.60	7.30	12.90	WDI
Female unemployment (%)	8.23	4.32	2.70	24.10	WDI
Female part-time emp. (%)	38.21	12.18	13.20	60.40	WDI
Life expectancy, female (years)	83.70	1.42	80.90	86.20	WDI
Life expectancy, male (years)	78.83	1.65	75.10	82.10	WDI

LFP = labour force participation; pp = percentage points. Data cover 14 Western European OECD countries (Austria, Belgium, Denmark, Finland, France, Germany, Ireland, Italy, Netherlands, Norway, Spain, Sweden, Switzerland, United Kingdom) from 2010–2022.